MILITARY SPECIFICATION

# SILVER ALLOY PELLET, DENTAL

This specification is approved for use by the Defense Personnel Support Center, Defense Supply Agency, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers alloys in pellet form, composed essentially of silver and tin, suitable for use in dentistry compounding amalgam.

1.2 Classification. The alloys covered by this specification shall be of the following types, as specified (see 6.1):

Type I. - Medium setting, 230 mesh.

Type II. - Fast setting, 325 mesh.

Type III. - Medium setting, Spherical, 270 mesh.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Standard For Testing Purposes.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, CODE ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

20555 - SILVER ALLOY PELLET (DENTAL) TESTER.

20776 - SILVER ALLOY PELLET CONTAINER AND DISPENSER.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN DENTAL ASSOCIATION (ADA)

Specification No. 1 Alloy for Dental Amalgam.

(Application for copies should be addressed to the Council on Dental Materials and Devices, American Dental Association, 211 East Chicago Avenue, Chicago, IL 60611.)



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3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of the silver alloy pellets shall conform to the requirements shown in table I, when tested as specified in 4.5.1.

TABLE I. Composition.

| Components | Type I, percent | Type II, percent | Type III, percent |
|---|---|---|---|
| Silver (minimum) | 68.0 | 69.6 | 68.0 |
| Tin (maximum) | 27.0 | 26.5 | 27.0 |
| Copper (maximum) | 5.0 | 3.9 | 5.0 |
| Zinc | 0.5-2.0 | 0.3-0.6 | 0.1-0.8 |

3.2 Working qualities. Type I and type II material shall be furnished in pellets compressed from silver alloy filings, and type III material shall be furnished in pellets compressed from spherical silver alloy, as specified herein. Silver alloy pellets shall provide a mass of amalgam suitable for restoration of natural teeth when triturated with mercury in a ratio recommended by the manufacturer.

3.2.1 Time of amalgamation. The amalgam made from a 2-pellet mix shall be ready for condensation no longer than 30 seconds from start of trituration when mechanically triturated, in accordance with the procedure as specified in 4.5.3.

3.2.2 Consistency. The alloy shall form a smooth plastic amalgam when tested as specified in 4.5.4.

3.2.3 Carving. The amalgam shall be susceptible to carving immediately after condensation and shall remain so for a maximum of 15 minutes after amalgamation, when tested as specified in 4.5.5.

3.3 Foreign material. The alloy shall be free of foreign material, when tested as specified in 4.5.12. The pellets shall be compressed without any binding agents or other inert ingredients for holding the pellets in shape.

3.4 Flow. Silver alloy specimens shall show no more than 3 percent flow (that is, no more than 3 percent shortening in length of specimen) within a period of 21 hours, when tested as specified in 4.5.6.

3.5 Dimensional change during hardening. The change in the length of specimens for all types alloy shall be 0 $\pm$ 20 microns per centimeter, at the end of 24 hours, when tested as specified in 4.5.7.



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3.6 Compressive strength. The minimum compressive strength at one hour shall be 14,000 pounds per square inch (p.s.i.) for type I alloy, and 18,000 p.s.i. for type II and type III, respectively, when tested as specified in 4.5.8.

3.7 Diametral tensile strength. The minimum diametral tensile strength shall be 500 p.s.i. for type I alloy and type III alloy and 800 p.s.i. for type II, when tested as specified in 4.5.9.

3.8 Cohesive time. The cohesive time of all alloys shall be a minimum of 3.5 minutes, when tested as specified in 4.5.10.

3.9 Manufacturer's instructions. Adequate and accurate instructions for manipulating and proportioning the alloy pellets shall be furnished with each 5-ounce unit package. Two copies of the instructions shall be forwarded to the Defense Personnel Support Center (ATTS-2) for information purposes. The following essential details shall be included in addition to any other necessary instructions.

3.9.1 Proportions. The ratio by weight of mercury to alloy, and the recommended plunger and number of spills for a single-pellet and for a 2-pellet mix, when using a calibrated mercury dispenser, NSN 6520-00-083-6461, shall be specified.

3.9.2 Mechanical trituration. The time setting on a mechanical amalgamator (NSN 6520-00-500-5010) for a single-pellet and two-pellet mix, using a plastic capsule and plastic pestle, NSN 6520-00-889-5727, shall be included. The instructions for mechanical trituration, furnished by the manufacturer, shall be of sufficient detail to permit reproducible test data when derived by specification test procedures.

3.9.3 Condensation. Details relating to condensation shall include when and how the excess mercury shall be expressed and the method of packing for small and large restorations.

3.9.4 Precautions. The following precautions shall be printed in type larger than that in the body of the instructions: "The alloy pellets contain zinc, and the amalgam made therefrom will show excessive corrosion and expansion if moisture is introduced during mixing and packing of the amalgam. Therefore, the amalgam should not be mulled in nor touched with the bare hands, nor condensed in a wet cavity, nor contaminated with moisture in any manner."

3.9.5 Finishing. Methods for finishing and polishing the amalgam restoration shall be included.

3.10 Finished pellet. The silver alloy shall be furnished in pellet form. Pellets, when triturated with mercury, shall form a smooth plastic mass (amalgam) suitable as a restorative dental material. The pellet shall be round in shape with a flat surface on the top and bottom faces. Pellets shall be



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of proper size to fit into, and dispense from the pellet container, when used with the pellet dispenser, NSN 6520-00-083-4556. Pellets shall be suitable for use with plastic capsule and plastic pestle, NSN 6520-00-889-5727.

3.11 Weight. The silver alloy pellets shall be furnished as a 5-ounce unit package. Each unit package of type I and II pellets shall consist of 5 individual 1-ounce containers. Each unit package of type III pellets shall consist of 4 individual 1-1/4-ounce containers. The individual containers shall conform to the weights specified in table II.

TABLE II. Alloy weights and tolerances.

| Type | Individual container | | | | Individual pellet | |
|---|---|---|---|---|---|---|
| | Weight | | Tolerance | | Weight | Tolerance |
| | Grams | Troy ozs. | Grams | Grains | Grains | Grains |
| I | 31.1 | 1.0 | +0.3 | +5 | 6.0 | ± 0.5 |
| II | 31.1 | 1.0 | +0.3 | +5 | 6.0 | ± 0.5 |
| III | 38.9 | 1.25 | +0.36 | +6 | 4.7 | ± 0.6 |

3.12 Pellet container. Each 1 ounce of pellets (type I and type II) and each 1-1/4 ounces of pellets (type III) shall be furnished in a plastic container with a slip-on cover. The cover shall be secured to the container by means of a suitable pressure sensitive label to prevent accidental opening or spillage of contents. The pellet container shall have 4 vertical columns properly spaced to provide positive dispensing of pellets in a pellet dispenser, NSN 6520-00-083-4556. (See Defense Personnel Support Center Drawing No. 20776.) The container shall be push fit into the dispenser, and alternate rows of pellets shall be positively dispensed without binding, breaking, or chipping.

3.12.1 Standard sample. Standard sample of pellet dispenser, NSN 6520-00-083-4556 is available at the Defense Personnel Support Center for determination of fit and function requirements. A sample of 3 pellet containers shall be drawn at random from each production lot and tested as specified in 4.5.16.

3.13 Lot or control number. Each individual container shall be legibly marked with the lot or control number. The size of each production lot or control shall be a maximum or 5,500 troy ounces.

3.14 Disintegration. The pellets shall completely break to a powder during the first 4 seconds of amalgamation, when tested as specified in 4.5.11.

3.15 Fineness. The fineness shall conform to the requirements specified in table III for the applicable type when tested as specified in 4.5.12.



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TABLE III. Fineness.

| | Percent minimum | Sieve number |
|---|---|---|
| Type I | 95.0 | 230 |
| Type II | 95.0 | 325 |
| Type III | 98.0 | 270 |
| | 78.0 | 400 |

3.16 Particle thickness. The particle thickness of the disintegrated pellet shall be no more than 0.0015 inch, when tested as specified in 4.5.13.

3.17 Hardness. Types I, II, and III alloy pellets shall break at no less than 3.0 kilograms, nor more than 4.5 kilograms. The average of 10 pellets shall be taken as the breaking value, and the range from the high value to the low shall be no more than 2.0 kilograms.

3.18 Certification. At the time of award, suppliers shall furnish a dated certificate and data showing that the brand of silver alloy pellet to be furnished is the same in quality and brand as that appearing on the list of certified materials for American Dental Association Specification No. 1 for Dental Amalgam Alloy. Compliance with this certification requirement does not absolve the supplier from complete compliance with the requirements of this specification.

3.19 Workmanship. The alloy pellets shall be first class in every respect. The pellets shall be clean and free from foreign particles. Pellets shall be free from all defects which either detract from their appearance or may affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.



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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables IV and VI (see 4.2.1 and 4.3) and table V (see 4.2.2).

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table IV. Unit of product for examination purposes shall be a 5-ounce package.

TABLE IV. Examination criteria.

| Type defect | Inspection level | AQL (percent defective) |
|---|---|---|
| Major A | II | 1.0 |
| Major B | S-2 | 1.5 |
| Minor | S-2 | 2.5 |

4.2.2 Sampling for tests. Sampling for tests shall be conducted on each lot or control number in accordance with table V. If any test determination for a lot does not comply with the requirements specified, the lot shall be rejected.

TABLE V. Test criteria.

| Characteristic | Requirement | Test procedure | Number of determinations per lot or batch number |
|---|---|---|---|
| Chemical composition | 3.1 | 4.5.1 | 2 |
| Time of amalgamation | 3.2.1 | 4.5.3 | 2 |
| Consistency | 3.2.2 | 4.5.4 | 2 |
| Carving | 3.2.3 | 4.5.5 | 2 |



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TABLE V. Test criteria (continued).

| Characteristic | Requirement | Test procedure | Number of determinations per lot or batch number |
|---|---|---|---|
| Foreign material | 3.3 | 4.5.12 | 1 |
| Flow | 3.4 | 4.5.6 | 2 /1 |
| Dimensional change during hardening | 3.5 | 4.5.7 | 2 /1 |
| Compressive strength | 3.6 | 4.5.8 | 1 /2 |
| Diametral tensile strength | 3.7 | 4.5.9 | 2 /2 |
| Cohesive time | 3.8 | 4.5.10 | 3 |
| Individual pellet weight | 3.11 | 4.5.15 | 24 |
| Net weight in 1 dispenser | 3.11 | 4.5.15 | see 4.5.11 |
| Disintegration | 3.14 | 4.5.11 | 15 |
| Fineness | 3.15 | 4.5.12 | 1 |
| Particle thickness | 3.16 | 4.5.13 | 1 |
| Hardness | 3.17 | 4.5.14 | 3 |
| Fit and function | 3.12.1 | 4.5.16 | see 3.12.1 |

/1 Each determination shall be the average of 2 specimens.

/2 Each determination shall be an average of 5 specimens.

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of 2 or more test characteristics.

4.2.4 Chemical composition. At least one 5-ounce package from each production lot or control (5,500 troy ounces, maximum) shall be selected for chemical analysis.

4.3 Examination. Silver alloy pellets shall be examined for defects including, but not limited to, those listed in table VI.

TABLE VI. Classification of defects.

| Category | Defect |
|---|---|
| Major A: | |
| 101 | Type not as specified. |
| 102 | Instructions for use are inadequate, incomplete, inaccurate, or missing altogether. |
| Major B: | |
| 151 | Pellets crushed or broken. 1/ |
| 152 | Finished pellets not as specified. 1/ |



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TABLE VI. Classification of defects (continued).

| Category | Defect |
|---|---|
| Minor: | |
| 201 | Pellet container design not as specified. |
| 202 | Finished pellet shape not as specified. |
| 203 | Lot or control number marking not present on pellet container. |

1/ Only 1 pellet container (selected at random from each unit package sampled) shall be examined.

4.4 Packaging inspection. The inspection of the unit package, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests.

4.5.1 Chemical analysis. The chemical composition shall be determined by a suitable, accurate wet or electrochemical analysis.

4.5.2 Preparation of specimens. Specimens shall be prepared by mechanically triturating, in accordance with manufacturer's instructions, 2 silver alloy pellets with the proper weight of mercury in proportion to the weight ratio specifed by the manufacturer. Specimens for compressive strength test shall be prepared as specifed in 4.5.8. The preparation of all test specimens and all tests shall be conducted at a temperature of 23.0 ± 2.0 degrees Centigrade (°C.) (73.4 ± 3.6 degrees Fahrenheit (°F.)), unless otherwise specified. In preparing specimens used for determining the diametral tensile strength, flow, dimensional change, and compressive strength, the mold shall be assembled as shown in figure 1 with plunger no. 1 with cap in position (used when dimensional change specimens are measured in an interferometer), or with plunger no. 3. The coherent mass of mechanically triturated amalgam shall be placed on top of the cavity of the die and shall be immediately inserted in the mold with several thrusts of an amalgam condenser that has a diameter slightly less than 4 millimeters. Mercury shall not be expressed during the insertion process. The no. 2 plunger shall be inserted and the time schedule in table VII shall be followed.



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TABLE VII. Schedule for preparation of specimens.

| Operation | Time (seconds) |
|---|---|
| End of trituration | 0 |
| Place triturated mass in mold and apply 14 meganewtons per square meter (2,030 p.s.i.) pressure | 30 |
| Release load and remove no. 2 spacer | 45 |
| Replace load | 50 |
| Release load | 90 |
| Brush away mercury and eject specimen. | 120 |
| Specimens shall not be trimmed. | |
| Store specimens for flow, dimensional change, and compressive strength at 37 ± 1 degree Centigrade. | |

4.5.3 Time of amalgamation. The time of amalgamation is the time which has elapsed between the beginning of the trituration and the time the amalgam is ready for packing. Two pellets shall be triturated as specified in 4.5.2, and the resulting amalgam (plastic mass) shall be transferred to a 3-inch-square rubber dam. The plastic mass shall be gently rolled (without mulling) in the rubber dam into a rope approximately 1 inch long. The amalgam shall be considered suitable for packing when the rope of amalgam retains a smooth homogeneous shape, having a shiny bright surface, without showing signs of crumbling.

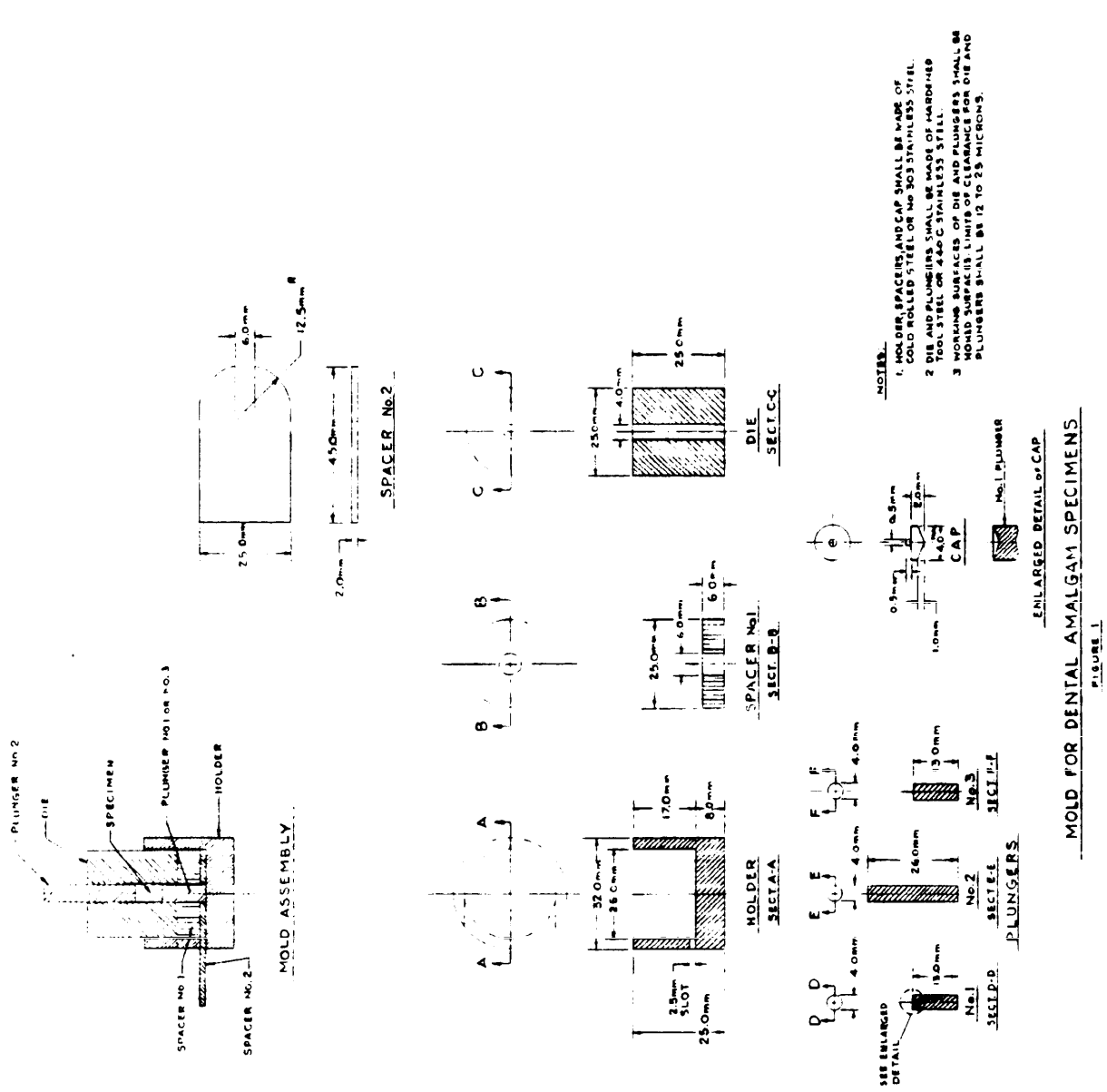
4.5.4 Consistency. Two pellets shall be triturated as specified in 4.5.2, and the resulting plastic mass shall be examined for smoothness by rolling the amalgam in the fingers. If the mass (amalgam) appears sandy or granular, it does not comply with requirements specified.

4.5.5 Carving. Practical tests using either mold dies or clinical testing shall be performed to determine the carving characteristics.

4.5.6 Flow. The specimen shall be prepared as specified in 4.5.2 using plungers no. 2 and 3 (see figure 1). The die shall be maintained at 37.0 ± 1.0° C. (98.6 ± 1.8° F.) except during the packing of specimen. The specimen shall be removed from the die on completion of condensation and shall then be transferred to an environment of 37.0 ± 1.0° C. (98.6 ± 1.8° F.). Prior to insertion in the flow micrometer, the ends shall be surfaced plane and at right angles to the axis. Three hours after amalgamation is started, the specimen shall be subjected to a constant axial load of 10.0 meganewtons per square meter (1,450 pounds per square inch). This load shall be maintained for 21 hours. The change in length of the specimen shall be determined for compliance with the requirements of paragraph 3.5. During this test the temperature of the specimen shall be maintained at 37.0 ± 1.0° C. (98.6 ± 1.8° F.). The average shortening obtained from 2 such specimens shall be the recorded value of flow for each determination.



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MOLD FOR DENTAL AMALGAM SPECIMENS

FIGURE 1



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4.5.7 Dimensional change during hardening. The specimen shall be prepared as specified in 4.5.2 using plungers no. 1 (with cap if an interferometer is used) and no. 2 (see figure 1). The die shall be maintained at 37.0 ± 1.0° C. (98.6 ± 1.8° F.) except during the packing of the specimen. The specimen shall be removed from the die as soon as condensation is completed and not later than 10 minutes from the start of mixing. The specimen shall then be transferred to an environment of 37.0 ± 1.0° C. (98.6 ± 1.8° F.). The specimen shall be placed in the measuring instrument and shall not be subjected to restraint during the test. The initial measurement shall be made 300 seconds (5 minutes) after the end of trituration. The final measurement shall be made at the end of 24 hours. During this test, the temperature of the specimen shall be maintained at 37.0 ± 1.0° C. (98.6 ± 1.8° F.). The average change in length of 2 such specimens shall be the recorded value for setting change for each determination.

4.5.8 Compressive strength. Five specimens shall be made. For Type I alloy, triturate 0.630 gms. silver alloy with 0.630 gms. mercury. For Type II alloy, triturate 0.630 gms. silver alloy with 0.739 gms. mercury. With Type III alloy, triturate 0.630 gms. alloy with 0.537 gms. mercury. Trituration is done in a plastic capsule with plastic pestle (NSN 6520-00-889-5727) in an amalgamator (NSN 6520-00-500-5010). Trituration times shall be as in table VIII.

TABLE VIII. Trituration times.

| Alloy | Time (seconds) |
|---|---|
| Type I | 16 |
| Type II | 20 |
| Type III | 20 |

Instructions for loading the die (4.5.2) and the time schedule in table VII of 4.5.2 shall be followed.

One hour from start of trituration, specimens shall be broken on an Instron Testing Machine using a crosshead speed of 0.010 inch per minute. Other suitable equipment may be used provided that prior approval is received from the contracting officer. The value for compressive strength shall be reported as the average of 3 or more specimens from a lot of 5 specimens and shall be rounded off to the nearest 100 pounds per square inch. If the values for individual specimens fall more than 15 percent below the average of the 5, they shall be discarded and the average of the remaining specimens shall be reported. In case more than 2 of the specimens are eliminated, the test shall be repeated.

4.5.9 Diametral tensile strength. Five specimens shall be prepared as specified in 4.5.2 using plungers no. 2 and 3 (see figure 1). The specimens shall be stored and the test shall be conducted at 23 ± 2 degrees Centigrade. The length and diameter of the specimens shall be measured with a hand micrometer, and the specimens shall be padded with 2 thicknesses of 0.038 millimeter (0.0015 inch) aluminum foil on each side. Each specimen shall be tested at 15 minutes



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from end of trituration. Specimens shall be fractured on an Instron Testing Machine. Other suitable equipment may be used provided that prior approval is received from the contracting officer. Compressive force shall be applied diametrically to the specimens using a loading speed of 0.50 millimeters per minute (0.02 inches per minute). The average value of 5 specimens shall be rounded to the nearest 0.1 meganewton per square meter (14.5 p.s.i.). If the values for individual specimens fall more than 15 percent below the average of the 5, they shall be discarded and the average of the remaining specimens shall be reported. In case more than 2 specimens are eliminated, the test shall be repeated. Tensile strength values shall be determined in accordance with the following equation:

$$St = \frac{2P}{\pi DL}$$

Where St = Tensile stress
P = Load at fracture
D = Diameter of specimen
L = Length of specimen

4.5.10 Cohesive time. Two pellets of alloy shall be placed in a plastic capsule with plastic pestle (NSN 6520-00-889-5727). Two spills of mercury shall be added using the specific or optimum plunger of the mercury dispenser (NSN 6520-00-083-4556), as stated in the manufacturer's instructions for the type of alloy involved.

4.5.11 Disintegration test. Two pellets shall be placed in a plastic capsule with plastic pestle, NSN 6520-00-889-5727, and triturated for 4 seconds in an amalgamator (NSN 6520-00-500-5010). For type I or type II alloy, the resulting 180 grains of alloy powder (15 determinations) shall be used for determining the foreign material and fineness, and particle thickness test specified in paragraphs 4.5.12 and 4.5.13, respectively. For type III alloy, the resulting 178 grains of alloy powder (19 determinations) shall be used for determining the foreign material and fineness, and particle thickness test specified in paragraphs 4.5.12 and 4.5.13 respectively.

4.5.12 Foreign material and fineness. Foreign material and fineness shall be determined by sieving a 10-gram portion of the disintegrated alloy tested in paragraph 4.5.11. Three-inch diameter sieves, with pans and covers attached, shall be alternately tapped and rotated 100 times per minute, while held in a slightly inclined position. Sieves shall conform to RR-S-366 and shall be No. 230 for type I alloy, No. 325 for type II alloy, and No. 270 and No. 400 for type III alloy. The operation shall be continued until less than 0.1 gram of the alloy passes through the sieve in 1 minute of continuous sieving. The amount of material retained on the sieve shall be weighed, and the fineness shall be calculated and expressed as a percentage of the original sample. In addition, any residue remaining on sieve shall be examined for foreign material at a magnification of 25X to 50X.



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4.5.13 Particle thickness test. The alloy used for determining the fineness shall be used for the particle thickness test. Measure the thickness of a pair of assembled and matched optical flats with a micrometer calibrated in 0.0001 inches. The 2 optical flats shall measure 1 by 1 by 1/8 inch and shall have 2 matching sides marked. Take 10 grams of type I, type II or type III silver alloy, as specified, and sieve through a No. 230 sieve (type I), No. 325 sieve (type II) or a No. 270 and No. 400 sieve (type III) using the test method descorbed in 4.5.12. Sieves shall conform to RR-S-366. Mix the silver alloy retained on the No. 230 sieve, on the No. 325 sieve, or on the No. 270 and No. 400 sieve thoroughly by shaking and turning in a 1-ounce bottle. If all the alloy passes through the No. 230 sieve, No. 325 sieve or No. 270 sieve, perform this test on the sieved-through alloy using that portion retained on the No. 400 sieve in the case of type III. Disperse a few particles of alloy on 1 optical flat, using the curved portion of an explorer so that no particle overlaps. Some particles should be present in each corner and some in the center. Place the second optical flat over the particles on the first flat and seat firmly. Clamp the 2 matched flats together with a 0.001 inch micrometer at the center of the assembly. Measure the thickness of the assembly at the 4 corners with another micrometer calibrated in 0.0001 inches. Average the 4 measurements and deduct the thickness obtained for the assembled pair of optical flats (withot alloy). All assemblies of the optical flats shall be made with the marked sides matching. Make 5 tests using fresh portions from the sample of powdered alloy. Report the average of the 5 tests as the particle thickness.

4.5.14 Hardness test. Pellets shall be tested in a Stokes hardness tester modified for use on alloy pellets as shown in DPSC Drawing No. 20555. A pellet shall be placed in the slot of the pellet holder so that the load of the plunger is applied axially to the edge of the pellet.

4.5.15 Weight. Weight shall be determined by means of a suitable capacity precision balance.

4.5.16 Fit and function. Insert pellet container into standard sample dispenser and actuate dispensing slide. Fit and function shall be as specified in 3.12.1.

## 5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue.

5.1.1.1 Type I and Type II. One package containing five individual one-ounce plastic dispensers of one type only, constitutes one unit of issue.



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5.1.1.2 Type III. One package containing four individual one and one quarter (1-1/4) ounce plastic pellet holders, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit (five plastic dispensers of Type I or Type II or four plastic holders of Type III shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.2.2 Intermediate package. Twelve unit packages containing silver alloy pellets of one type only shall be packaged in a box conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.1.3 Level C. Each unit (five plastic dispensers) of Type I or Type II or four holders of Type III shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Six intermediate packages containing silver alloy pellets of one type only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. In addition, each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636. Wood boxes shall be lined with case liners conforming to MIL-L-10547. Case liners shall be closed and sealed as specified in the liner specification.

5.2.2 Level B. Six intermediate packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The packaged pellets shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.



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5.3 Marking.

5.3.1 Individual plastic containers. Each plastic container shall be labeled with the item identification, lot (control) number and the legend: "Fast Setting" for Type II and "Medium Setting" for Types I and III.

5.3.2 Interior packages. Each unit and intermediate package shall be marked as specified in MIL-STD-129. Lot (control) number, contract or purchase order number, and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked as specified in MIL-STD-129. Lot (control) number shall be shown.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specificaion.
(b) Type of alloy required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Supersession data. This specification includes the requirements of Purchase Description No. 2 dated 19 July 1973.

6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

| National Stock Number | Item Identification | Specification Designation |
|---|---|---|
| 6520-00-080-1651 | SILVER ALLOY PELLET, DENTAL, Medium Setting, Spherical, 270 Mesh, 512s, 5 oz. | Type III |
| 6520-00-853-8346 | SILVER ALLOY PELLET, DENTAL, Medium Setting, Fine-Grain, 230 Mesh, 400s, 5 oz. | Type I |
| 6520-00-926-2113 | SILVER ALLOY PELLET, DENTAL, Fast Setting, Micro-Grain, 325 Mesh, 400s, 5 oz. | Type II |

Preparing activity:
DSA-DM

Project No. 6520-1452



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers, vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-S-37709 (DSA-DM) SILVER ALLOY PELLET, DENTAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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